Quay Installation using Mirror-Registry

Download the Mirror Registry version 1.3.9 installation package from the official Red Hat developer repository to your local system.

```
wget https://developers.redhat.com/content-gateway/file/pub/openshift-v4/clients/mirror-registry/1.3.9/mirror-registry.tar.gz
```

Untar the file

```
tar xvf mirror-registry.tar.gz
```

Generates a new 2048-bit RSA private key and saves it as a file named redhat.key, which is typically used for creating SSL/TLS certificates or securing communications.

```
openssl genrsa -out redhat.key 2048
```

Generate a CSR file (redhat.csr) using the private key redhat.key for SSL certificate creation.

```
openssl req -key redhat.key -new -out redhat.csr
```

```
[root@grafana quay_ins]# openssl genrsa -des3 -out redhat.key 2048
Enter PEM pass phrase:
Verifying - Enter PEM pass phrase:
[root@grafana quay_ins]# ls
execution-environment.tar  image-archive.tar  mirror-registry  mirror-registry.tar.gz  redhat.key
[root@grafana quay_ins]# openssl req -key redhat.key -new -out redhat.csr
Enter pass phrase for redhat.key:
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [XX]:IN
State or Province Name (full name) []:karnataka
Locality Name (eg, city) [Default City]:bengaluru
Organization Name (eg, company) [Default Company Ltd]:example
Organizational Unit Name (eg, section) []:testing
Common Name (eg, your name or your server's hostname) []:redhat
Email Address []:jayarajvp2005@gmail.com

Please enter the following 'extra' attributes
to be sent with your certificate request
A challenge password []:Redhat@123
An optional company name []:non
[root@grafana quay_ins]#
```

Create a self-signed SSL certificate (redhat.crt) valid for 90 days, using the CSR (redhat.csr), private key (redhat.key), and SAN settings from openssl-san.cnf.

```
openssl x509 -req -in redhat.csr -signkey redhat.key -out redhat.crt -days
90 -extensions req_ext -extfile openssl-san.cnf
```

```
[root@grafana quay_ins]# openssl x509 -signkey redhat.key -in redhat.csr -req -days 90 -out redhat.crt
Enter pass phrase for redhat.key:
Certificate request self-signature ok
subject=C=IN, ST=karnataka, L=bengaluru, O=example, OU=testing, CN=redhat, emailAddress=jayarajvp2005@gmail.com
[root@grafana quay_ins]# ls
execution-environment.tar  image-archive.tar  mirror-registry  mirror-registry.tar.gz  redhat.crt  redhat.csr  redhat.key
[root@grafana quay_ins]# cat redhat.crt
-----BEGIN CERTIFICATE-----
MIIDlTCCAr2gAwIBAgIUBJeSFP3eyN+DMZxxnoFXhPYHsywwDQYJKoZIhvcNAQEL
BQAwgZIxCzAJBgNVBAYTAklOMRIwEAYDVQQIDAlrYXJuYXRha2ExEjAQBgNVBAcM
CWJlbmdhbHVydTEQMA4GAlUECgwHZXhhbXBsZTEQMA4GAlUECwwHdGVzdGluZzEP
MA0GAlUEAwwGcmVkaGF0MSYwJAYJKoZIhvcNAQkBFhdqYXlhcmFqdnAyMDAlQGdt
YWlsLmNvbTAeFw0yNTExMTAwNTMwMTlaFw0yNjAyMDgwNTMwMTlaMIGSMQswCQYD
VQQGEwJJTjESMBAGAlUECAwJa2FybmF0YWthMRIwEAYDVQQHDAliZW5nYWxlcnUx
EDAOBgNVBAoMB2V4YW1wbGUxEDAOBgNVBAsMB3Rlc3RpbmcxDzANBgNVBAMMBnJl
EGhhdEEmMCQGCSqGSIb3DQEJARYXamF5YXJhanEwMjAwNUBnbWFpbC5jb20wggEi
MA0GCSqGSIb3DQEBAQUAA4IBDwAwggEKAoIBAQC704TScClaUceug32gPRQ/EaRW
CPBEFHfOiSOwnBKvjjl+Oih1TvPvJTQSWCwYu7o40cHY9zLZ5oSETyhAolNl1EV3
lRJ9LBcshTk6al75K3sG6EPN6yhtrtkiPr362oVYnBO0AWRdnZl+BWj4mammjfZR
7PchMDFttJMzGjOKfuIqhm1I8fSLEu/AfGza2V2izVZgpEdlN+CVWnQtUeOqqsaa
vI54xH4th5Cp50gVObRZwCKa3DenaQUlWFgASQlg5rd9Id6RjS2KJ0ukg/48gkCi
o7xmmunTtM2XsAnmtbg6St5qgrI45HemmSLFmoEQyKF3u2A+0vXUIeWC1ybhAgMB
AAGjITAfMB0GAlUdDgQWBBRsxQ1LpYepgJ4It2JkjxM4ADYLdTANBgkqhkiG9w0B
AQsFAAOCAQEAORxtGDOMFBFXO2BKXtu6mj51nq4WsJdVM9SV79lSokqY+HPd+Rqr
hi+AdY6Qo8YqpAW5Dr65CN3MYolrrkSgBj5S5Gq2N9m7WH5bPaRjT6d2qJkHaHDh
p1EdsJ16ouawo0OIWGbYHv33Ejq/GBFBtV/hUAcAXUQ6h1501DMRL+1HSCp4f4sV
80mjBTjKmnm2k+X9T+376e11BS3/it1rv8Skrjbu8mtavK1eRr10+2zp3Fb42rpz
qLAQMj+VApKJiWv9NaGLiAPrqRgSK+b/4xhJv7onTBSAQDILCfT/r1mBO+MehIml
uuxJSQomXxbXINC1dqiAEmoo/LveZhMXGA==
-----END CERTIFICATE-----
```

Install the Red Hat Quay Mirror Registry, setting its hostname & admin password and securing it with the provided SSL certificate (redhat.crt) and private key (redhat.key).

```
./mirror-registry install --quayHostname grafana.example.lab --initPassword
Redhat@123 --sslCert /root/quay_ins/cert/redhat.crt --sslKey
/root/quay_ins/cert/redhat.key
```

Currently logged in as superadmin

Certificate Viewer: grafana.example.lab

General Details

Issued To

Common Name (CN)	grafana.example.lab
Organization (O)	ExampleLab
Organizational Unit (OU)	<Not Part Of Certificate>

Issued By

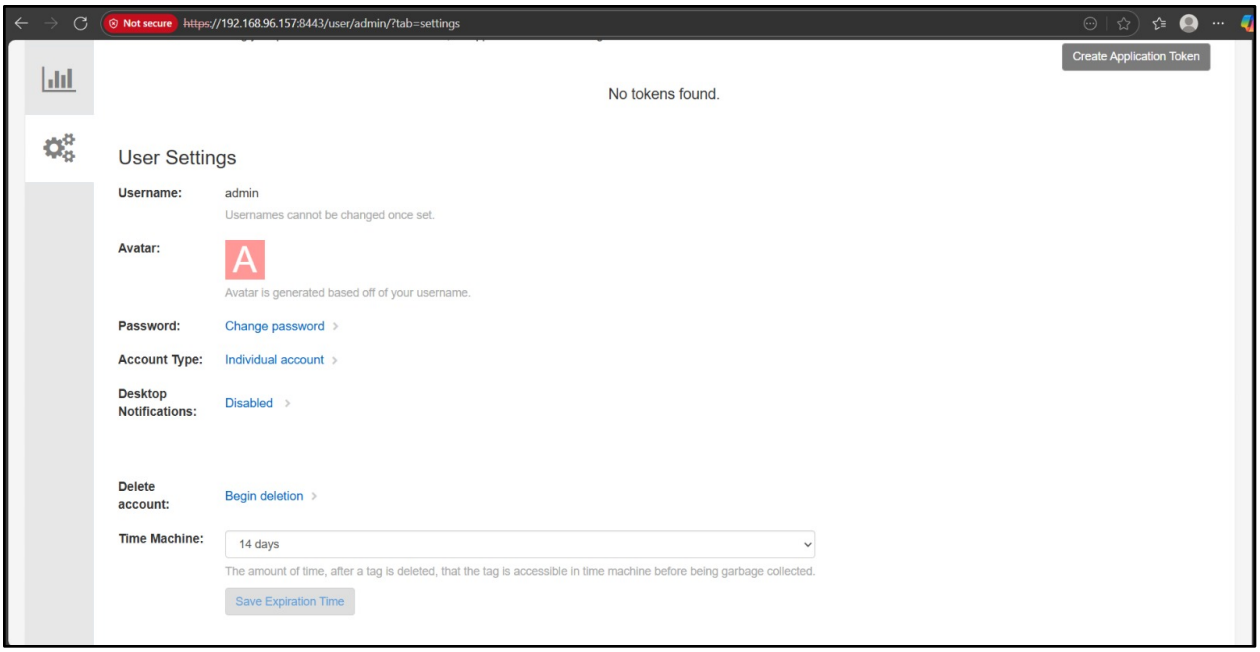
Common Name (CN)	grafana.example.lab
Organization (O)	ExampleLab
Organizational Unit (OU)	<Not Part Of Certificate>

Validity Period

Issued On	Monday, November 10, 2025 at 11:52:16 AM
Expires On	Sunday, February 8, 2026 at 11:52:16 AM

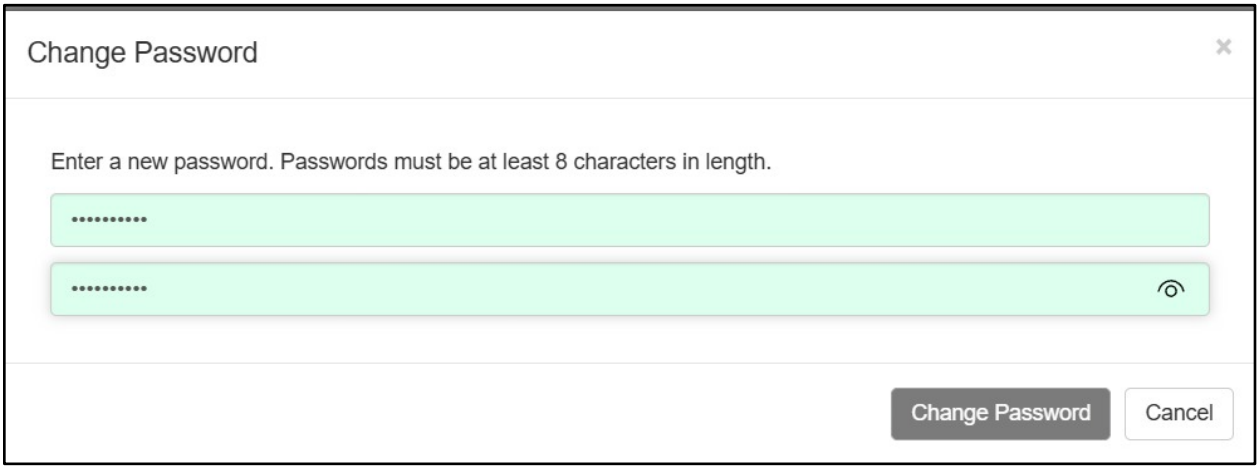
SHA-256 Fingerprints

Certificate	a31448f50d311348bce9848132528d54f10a241145f9ba522e8257c61c5f61b7
Public Key	c08996b563b443ed08b51813803c16d9e3c7e3bca4e2e80ce34311e5be8fe420



Changing the password for superadmin

Method 1



Method 2

Using CLI

```
pip3 install passlib  #password libraries installing
pip3 install bcrypt
```

```
python3 - <<'PY'
```

```
import bcrypt
pw = b"NewStrongP@ssw0rd!"      # keep <72 bytes
print(bcrypt.hashpw(pw, bcrypt.gensalt(rounds=12)).decode())
PY
```

```
UPDATE public."user"
SET password_hash =
'$2b$12$S8nKySgR6hlEjrlKgqe9Y.MHCJFn/fdEGuS1n8yeHQ.wWYtTDbZjW'
WHERE username = 'admin';
```

Replacing SSL Certificate

Creates a self-signed SSL certificate (redhat.crt) valid for 91 days using the CSR, private key, and SAN config.

```
openssl x509 -req -in redhat.csr -signkey redhat.key -out redhat.crt -days
91 -extensions req_ext -extfile openssl-san.cnf
```

Navigates to the Quay configuration directory.

```
cd /root/quay-install/quay-config
```

Removes the old SSL certificate and key files.

```
rm ssl.cert ssl.key
```

Moves the newly created key and CSR files to the current directory.

```
mv redhat.key redhat.csr .
```

Renames redhat.key to ssl.key for Quay to use as its SSL private key.

```
mv redhat.key ssl.key
```

Renames redhat.csr to ssl.cert so Quay uses it as the SSL certificate file.

```
mv redhat.csr ssl.cert
```